

NETWORK TECHNOLOGY	
Project Type	Specific Criteria
The system, application, or algorithm developed is related to one of the fields below: <u>Network Protocol</u> Example: Data Efficiency Rate for Indoor LiFi <u>Internet of Things</u> Example: Internet of Things for Home Invasion Systems <u>Network Security</u> Example: Security Operations Center <u>Ad Hoc Network</u> Example: Routing Protocol for Ad hoc Networks <u>Sensor Network</u> Example: Patient Tracking System <u>Blockchain</u> Example: Secure Voting System based on Blockchain <u>Cloud Computing</u> Example: Cloud based Attendance System <u>Cryptography</u> Example: Lightweight Cryptography Performance Rate	Project development must meet criteria based on the type of project as follows: - Can build simulations for network protocols - Test and then analyze the results for comparison between two network protocols. - Using different parameters to produce new results. - Using a new situation for a newly built application. - Test and then analyze the results for a built protocol. - Construction of network applications, protocols, and wireless applications – for example, ZigBee. - Successfully fully developed the built application. - Analyze protocol usage in depth by way of code usage and application construction - Use of existing communication tools, techniques, and protocols – examples of real-world use of zigbee and WiFi .

SUBMISSION DOCUMENT GUIDELINES

D1 - PROJECT PLANNING

INTRODUCTION

- Introduce the title by briefly explaining the main points that led to the identification of the problem/requirement.
- State the importance of the title by providing background and appropriate information (i.e., using facts and citing sources).

PROBLEM STATEMENT

- Clearly define the problem the project will address (i.e. a clear statement of the problem/need you have identified to address).
- Describe the problem.

PROPOSED SOLUTIONS

Based on the problem that has been identified, explain the possible solutions that the project will produce.

OBJECTIVE

State the project objectives based on the SMART approach:

- ✓ S – Specific
- ✓ M – Can be measured (Measurable)
- ✓ A – Achievable
- ✓ R – Relevant
- ✓ T – Duration (Timed)

SCOPE

Specify the scope of the final product (i.e. what will be included and what will not be included).

RESTRICTIONS

State the conditions, circumstances or events that may prevent and limit the implementation of the project (e.g. domain complexity, data confidentiality, access to experts and others).

METHOD

- Propose and explain the specific development process model used in the project (e.g. Waterfall, Increment Development, Spiral, Reuse/Component-oriented, Rational Unified Process).
- Explain why the process model was chosen.

IMPLEMENTATION SCHEDULE

- State the tasks/activities to be implemented as a Work Breakdown Structure (WBS).
- Identify the relationship between activities.
- Estimate the time required to perform each activity.
- Identify project milestones.
- Organize activities using a Gantt Chart (which shows the key relationships and tracks of activities).

SUMMARY

- Give a summary of the contents of D1.
- Emphasize the importance of the project.

REFERENCE

- List sources of information that support the proposal (ie books, newspaper clippings, conference papers and journal articles, web portals, etc.).
- Use the UKM Style reference format.

D2 - LITERATURE REVIEW

INTRODUCTION

- Introduce the topic of the study by giving general overview of the field of schedule.
- Explain the importance of the topic and the reason for choosing it.

CURRENT RESEARCH AND RELATED TECHNOLOGIES

- Review of Past Studies: State the key studies conducted in the same or related field. Include the main findings of these studies relevant to the technology or issues you are investigating.
- Existing Systems or Technologies: Analyze several systems or technologies currently used in the industry or ongoing research related to your project.
- Current Trends and Developments: Focus on studies examining the latest technological advancements and how these technologies have impacted the field being studied.

METHODOLOGY IN PAST STUDIES

- Research Approaches Used: Discuss the research methods or technical approaches employed by previous researchers, such as algorithms, mathematical models, or technology platforms.
- Effectiveness and Weaknesses of Methodologies: Identify the strengths and weaknesses of the methodologies used in previous studies. Explain how your research may improve upon or utilize similar methods.

COMPARISON AND CRITIQUE OF PAST STUDIES

- Comparison of Technologies: Compare various technologies or systems discussed in previous studies. Discuss the similarities and differences between the technologies used, including performance, cost, or ease of use.
- Critique of Past Studies: Provide a critical analysis of the shortcomings or limitations of previous studies, such as result accuracy, data deficiencies, or sample limitations.

RESEARCH GAPS

- Research Gaps: Identify gaps in the existing research that have not been explored. This may involve aspects of technology that have not been enhanced or specific constraints that remain unresolved.
- Importance of Your Study: Explain how your project aims to fill these gaps or provide solutions to unresolved problems.

INNOVATIVE SOLUTIONS IN PAST STUDIES

- **Innovative Approaches:** Discuss the new solutions or approaches proposed in previous studies and how these solutions inspire your project.
- **Recommendations for Improvement:** Suggest ways to enhance or modify existing technologies or methodologies based on the findings from previous studies.

SUMMARY

- **Summary of Key Findings:** Summarize the main results from previous studies and their relevance to your project.
- **Rationale for the Study:** Explain how the findings from this review support the rationale for conducting your final year project and contribute to technological advancements.

Writing Tips:

- **Critical and Analytical:** Ensure you critically evaluate previous studies rather than merely listing conducted research.
- **Use Up-to-Date Sources:** Include recent studies and technologies to ensure the relevance of your literature review (recent 5 years)
- **Reliable Sources:** Refer to academic journals, books, and recognized research reports in technology.
- **Use Clear Structure:** Organize the literature review into clear subtopics to facilitate reading and understanding.

REFERENCE

- List sources of information that support the proposal (i.e. books, newspaper clippings, conference papers and journal articles, web portals, etc.).
- Use the GAYA UKM reference format.

*Include figures or tables related to this topic.

D3 - REQUIREMENTS ANALYSIS AND SPECIFICATION

INTRODUCTION

Explain how the proposed solution fits the overall strategic or business objectives of an organization/community (i.e. how it will work with other systems or environments) or fits the identified problem based on the approach of the respective program area (for example, artificial intelligence, networks, information science and others).

DEFINITION OF USER NEEDS

Describe the services provided for users (functional needs of users) using clear language that is easy to understand.

SYSTEM REQUIREMENT SPECIFICATIONS*

- Specify system functional requirements for each user functional requirement.
- Specify non-functional (quality) requirements and domain requirements.
- Determine hardware and software requirements during development (for developers to build the product) and deployment (for users to use the product).
- For projects in the network field, specify the system requirements through the OSI model to specify the protocols involved in network communication or specify the cyber security requirements found in the developed system.

SYSTEM MODEL*

Choose **ONE** of the following approaches: 1) Object-oriented approach OR 2) structured analysis; **DO NOT combine the two.**

1. Object-oriented approach:

- Case diagrams and use case specifications to describe the functional requirements of the system.
- A sequence diagram shows the interactions between the user and the system.
- Activity diagram or flow chart to show the process steps in sequence.
- For projects in the field of networking, data communication diagrams through the OSI model.

OR

2. Structured analysis approach:

- A context diagram shows the scope and boundaries of the system.

- Flow diagram to show the level of detail of the process (at least up to the 2nd level).
- A flow chart shows the sequence of the steps of the process.

SUMMARY

- Summarize the content of D3.

Note:

* This is a guide. It may not fit your project. You are asked to discuss with the supervisor or Program Leader to come up with the exact specifications and model of the system.

REFERENCE

- List sources of information that support the proposal (i.e. books, newspaper clippings, conference papers and journal articles, web portals, etc.).
- Use the GAYA UKM reference format.

APPENDIX

- Briefly explain the data selection method for product requirements (who, when, what, why, how). Include some evidence (examples of interview questions / scripts, survey questionnaires, photos as evidence of observations, etc.).
- Briefly describe the requirements validation process (such as testing, reviews / walkthroughs, and prototyping).

D4 - DESIGN SPECIFICATION

INTRODUCTION

Give an overview of the document:

- Describe the purpose of this document.
- Describe the scope of this document.
- Link this document with other related documents:
 - ✓ Documents that provide background and/or context for this document (e.g. Project Plan/Proposal, Requirements Specification).
- Define any important terms, acronyms or abbreviations.

ARCHITECTURAL DESIGN

- Identify and explain the appropriate architecture design to use to develop the system (e.g. Layered, Model-View-Controller, Repository, Client-Server, Pipe-Filter and others)
- Describe the modules/sub-systems and their relationships using appropriate diagrams (e.g. Module Hierarchy Chart/Structure Chart, etc.).
- For projects in the network field, identify and explain the network topology design suitable for developing the system (Logical and physical topology - hardware, MAC address, IP address and so on).

DATABASE DESIGN (If relevant)

- Class Diagram (for object-oriented approach) OR Entity Relationship Diagram (for structured analysis approach).
- Data Dictionary is used to list the attributes of each entity OR class.

ALGORITHM DESIGN

- Design the algorithm to provide system functions (for example, pseudo code, decision table / tree, system / program flow chart).
- Describe the behavior of the system, if any (such as a Statechart diagram or a Sequence diagram of the interaction between objects for an object-oriented approach).

INTERFACE DESIGN

- Sketch user interfaces using low-fidelity (i.e. card-based prototyping) or high-fidelity prototyping (like Macromedia Director / Visual Basic / MS PowerPoint etc.).
- Design system interfaces between modules/sub-systems.
- Storyboard to illustrate the UI / UX of a multimedia system.

Note: The design required depends on the proposed solution and the respective field. Use as needed. Consult the Program Leader for more information.

SUMMARY

- Summarize the contents of D4.

REFERENCE

- List sources of information that support the proposal (i.e. books, newspaper clippings, conference papers and journal articles, web portals, etc.).
- Use the UKM Style reference format.

D5 - PROPOSAL REPORT

ABSTRACT

Provide a brief overview of the entire project (in one paragraph), containing the following information:

- The main aspect/object/subject of the project.
- The problem faced by the aspect/object/subject is to be solved.
- Suggested solutions to overcome the stated problem.
- The methodology used to develop/produce the proposed solution.
- Expected project results.

CHAPTER 1: PROJECT PLANNING {taken from D1}

- **Introduction:** Introduce the project by briefly explaining the main points that led to the identification of the problem/requirement. Demonstrate the importance of the project by providing appropriate information or background.
- **Problem Statement:** Clearly state the specific problem the project will address. Describe the problem.
- **Proposed Solution:** Based on the problem that has been identified, explain the possible solution that the project will produce. Explain how the proposed solution fits the overall strategic or business objectives of an organization/community (i.e. how it will work with other systems or environments) or fits the identified problem based on the approach of the respective fields (for example, artificial intelligence, networks, information science and others).
- **Objectives:** State the objectives of the project based on the SMART approach.
- **Scope:** Specify the scope of the final product (i.e. what will be included and what will not be included).
- **Constraints:** State the conditions, circumstances or events that can prevent and limit the implementation of the project (e.g. domain complexity, data confidentiality, access to experts and others).
- **Method:** Describe the specific approach or development process model used in the project (e.g. Waterfall, Increment Development, Spiral, Reuse/Component-oriented, Rational Unified Process). State why such a model or approach was chosen.
- **Implementation Schedule:** State the tasks/activities to be implemented as a Work Breakdown Structure (WBS). Identify relationships between activities. Estimate the time required to perform each activity. Identify project milestones. Organize activities using a Gantt Chart (which shows the key relationships and tracks of activities).

CHAPTER 2: LITERATURE REVIEW {taken from D2}

- **Introduction:** Introduce the chapter.
- **[Chapter content]:** Describe existing systems/technologies/approaches relevant to your project. Discuss the system/technology/approach. Compare and contrast existing systems/technologies/approaches and provide proposed solutions.
- **Conclusion:** Give a summary of the chapter.

CHAPTER 3: METHODOLOGY {taken from D3 and D4}

- **Introduction:** Introduce the chapter.
- **User Needs:** Describe the service provided for the user (functional user needs) using language that is easy to understand.
- **System Requirements:** Specify system functional requirements for each user functional requirement. Specify non-functional (quality) requirements and domain requirements. Determine hardware and software requirements during development (for developers to build the product) and deployment (for users to use the product). For projects in the network field, specify the system requirements through the OSI model to specify the protocols involved in network communication or specify the cyber security requirements found in the developed system.
- **System Model:** Explain the system model either using 1) Object-oriented approach OR 2) structured analysis; DO NOT combine the two.
- **Architecture:** Identify and explain the appropriate architectural design to develop the system (eg *Layered*, *Repository*, *Model-View-Controller*, *Client-Server*, *Pipe-Filter* and others). Describe the modules/sub-systems and their relationships using appropriate diagrams (eg Module Hierarchy Chart/Structure Chart, etc.). For projects in the network field, identify and explain the network topology design suitable for developing the system (Logical and physical topology - hardware, MAC address, IP address and so on).
- **Database:** Database design either uses a Class Diagram (for an object-oriented approach) OR an Entity Relationship Diagram (for a structured analysis approach). Set up a Data Dictionary to list the attributes of each entity OR class.
- **Algorithms:** Design algorithms used to provide system functionality (e.g. pseudo code, decision tables/trees, system/program flow charts). Describe the behavior of the system, if any (such as a Statechart diagram or a Sequence diagram of the interaction between objects for an object-oriented approach).
- **Interface:** Sketch the user interface using low fidelity (i.e. card-based prototyping) or high-fidelity prototyping (such as *Macromedia Director/Visual Basic/MS PowerPoint* and others). Design system interfaces between modules/sub-systems. Storyboard to illustrate the UI/UX of a multimedia system.
- **Conclusion:** Give a summary of the chapter.

REFERENCE

- Recent and diverse references related to the project (books, newspaper clippings, conference papers and journal articles, web portals, etc.).
 - Five (5) years ago
 - Minimum of 10 references
 - Journal reference
- Use the GAYA UKM reference format.

APPENDIX

- Any document or material that can support the project proposal.

Note*:

- i. Number of Pages: Minimum - 30 pages, Maximum - 50 pages.
- ii. The content of these chapters depends on the proposed solutions and their respective fields. Use as needed. Consult the Program Leader for more information.

EVALUATION RUBRIC

SCORES DISTRIBUTION

The evaluation of TTT_4172 is divided into two, namely;

1. **Continuous assessment** represents 40% of the mark and is assessed by the project supervisor.
2. **The final assessment** represents 60% of the mark and is assessed by the supervisor and examiner at the semester's end (proposal presentation and proposal report).

Tables 1 through 5 display the distribution and scoring structure.

Table 1: Score distribution based on learning domains

Learning Domain	Continuous (40%)		Final – Pra-KID (60%)		Total	Distribution of marks	
	Supervisor	Examiner	Supervisor	Examiner		Supervisor	Examiner
Affective (A)	20		5	10	35	25	10
Cognitive (K)	20		15	30	65	35	30
Total	40		20	40	100	60	40

Table 2: Score distribution based on indicators

Domain Learning	Supervisor		Examiner	
	Indicator	Per cent	Indicator	Per cent
Affective (A)	Meeting 1 & 2 - D1 and D2 (Assessment 1 - 6%)	25%	Presentation – Examiner Form	5%
	Meeting 3 - D3 (Assessment 2 - 7%)		Demo of the system/product design – Examiner Form	5%
	Meeting 4 - D4 (Assessment 3 - 7%)			
	Meeting 5 & 6 – D5 (Pra-KID Assessment - 5%)			
Cognitive (K)	Meeting 1 & 2 – D1 and D2 (Assessment 1 8%)	35%	Report (D5) – Examiner Form	30%
	Meeting 3 – D3 (Assessment 2 - 5%)			
	Meeting 4 – D4 (Assessment 3 - 7%)			
	Meeting 5 & 6 – D5 (Pra-Kid Assessment - 15%)			
	Total	60%	Total	40%

Table 3: Rubric of Continuous Assessment

EVALUATION 1		
D1: PROJECT PLANNING and D2 : LITERATURE REVIEW (14%)		
Supervision and Presentation (K4% + A4%)	Contents of D1 and D2 (K4%)	Commitment (A2%)
0. No presentation took place.	0. No report.	0. Students do not show any commitment.
1. The presentation was ineffective.	1. The writing of the report is unclear or does not discuss the main ideas required.	1. Students show very low commitment.
2. Presentation is less effective.	2. The writing of the prepared report is not clear.	2. Students show low commitment.
3. The presentation was quite effective.	3. The report writing contains some fairly clear information.	3. Students show moderate commitment.
4. Effective presentation.	4. Report writing contains clear information.	4. Students show good commitment.
5. The presentation of information is explained very effectively.	5. The writing of the report contains very clear and sufficient information.	5. Students show high commitment.
EVALUATION 2		
D3: REQUIREMENTS ANALYSIS AND SPECIFICATION (12%)		
Supervision and Presentation (K2% + A4%)	Contents of D3 (K3%)	Commitment (A3%)
0. No presentation took place.	0. No report.	0. Students do not show any commitment.
1. The presentation was ineffective.	1. The writing of the report is unclear or does not discuss the main ideas required.	1. Students show very low commitment.
2. Presentation is less effective.	2. The writing of the prepared report is not clear.	2. Students show low commitment.
3. The presentation was quite effective.	3. The report writing contains some fairly clear information.	3. Students show moderate commitment.
4. Effective presentation.	4. Report writing contains clear information.	4. Students show good commitment.
5. The presentation of information is explained very effectively.	5. The writing of the report contains very clear and sufficient information.	5. Students show high commitment.
EVALUATION 3		
D4: DESIGN SPECIFICATIONS (14%)		

Supervision and Presentation (K2% + A4%)	Contents of D4 (K5%)	Commitment (A3%)
0. No presentation took place.	0. No report.	0. Students do not show any commitment.
1. The presentation was ineffective.	1. The writing of the report is unclear or does not discuss the main ideas required.	1. Students show very low commitment.
2. Presentation is less effective.	2. The writing of the prepared report is not clear.	2. Students show low commitment.
3. The presentation was quite effective.	3. The report writing contains some fairly clear information.	3. Students show moderate commitment.
4. Effective presentation.	4. Report writing contains clear information.	4. Students show good commitment.
5. The presentation of information is explained very effectively.	5. The writing of the report contains very clear and sufficient information.	5. Students show high commitment.